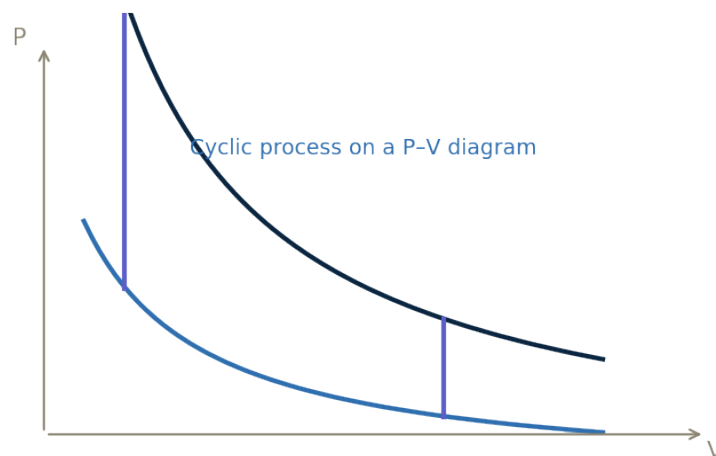


Thermodynamics

JEE Physics — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- The First Law of Thermodynamics: $\Delta Q = \Delta U + \Delta W$ (heat supplied = change in internal energy + work done by the system).
- For an ideal gas, internal energy depends only on temperature: $\Delta U = n C_v \Delta T$.
- Isothermal process: T constant, $\Delta U = 0$, so $\Delta Q = \Delta W = nRT \ln(V_2/V_1)$.
- Adiabatic process: $\Delta Q = 0$, so $\Delta U = -\Delta W$. Follows $PV^\gamma = \text{constant}$.
- Isobaric process: P constant, $W = P \Delta V$. Isochoric (isovolumetric) process: V constant, $W = 0$, so $\Delta Q = \Delta U$.
- Efficiency of a heat engine: $\eta = W(\text{output})/Q(\text{input}) = 1 - Q_2/Q_1$. Carnot engine efficiency: $\eta = 1 - T_2/T_1$ (temperatures in Kelvin).
- Second Law of Thermodynamics: heat cannot spontaneously flow from a colder to a hotter body without external work (Clausius statement); no engine can convert heat completely into work (Kelvin-Planck statement).
- $C_p - C_v = R$ (Mayer's relation) for an ideal gas; $\gamma = C_p/C_v$.



A cyclic thermodynamic process shown on a P-V diagram.

2. Formula Sheet

First law of thermodynamics	$\Delta Q = \Delta U + \Delta W$
Work in isothermal process	$W = nRT \ln(V_2/V_1)$
Adiabatic relation	$P V^\gamma = \text{constant}$
Work in isobaric process	$W = P (V_2 - V_1)$
Carnot efficiency	$\eta = 1 - T_2/T_1$
Mayer's relation	$C_p - C_v = R$
Internal energy (ideal gas)	$U = n C_v T$
Heat engine efficiency	$\eta = W/Q_1 = 1 - Q_2/Q_1$

3. Practice Problems (22 questions)

1. In a process, 500 J of heat is supplied to a gas and it does 200 J of work. Find the change in internal energy.
2. A gas expands isothermally from 2 L to 6 L at 300 K. Find the work done ($R = 8.31 \text{ J/mol K}$, $n = 1 \text{ mol}$).
3. A Carnot engine operates between 500 K and 300 K. Find its efficiency.
4. A gas undergoes an isochoric process. State whether work is done by the gas, and explain why.
5. A Carnot engine absorbs 800 J of heat from a hot reservoir at 400 K and rejects heat to a cold reservoir at 300 K. Find the work done by the engine.
6. State the first law of thermodynamics and explain the sign convention used for ΔQ and ΔW .
7. For a monatomic ideal gas, $C_v = (3/2)R$. Find C_p and γ .
8. A gas is compressed adiabatically, doing -300 J of work on it. Find the change in internal energy of the gas.
9. Explain why the temperature of a gas increases during adiabatic compression.
10. One mole of an ideal gas at pressure $2 \times 10^5 \text{ Pa}$ expands at constant pressure from a volume of 1 L to 3 L. Find the work done by the gas.
11. State the Kelvin-Planck statement of the second law of thermodynamics.
12. A refrigerator extracts heat from a cold body and rejects it to a hot body. Explain why this requires external work input, referencing the second law.
13. A Carnot refrigerator works between 260 K and 300 K. Find its coefficient of performance.
14. Explain the difference between an isothermal and an adiabatic process for an ideal gas.
15. In a cyclic process, the net change in internal energy is zero. If the gas absorbs 600 J of heat, find the net work done by the gas over one cycle.
16. Define a reversible process and give one example (idealized) of such a process.
17. For a diatomic ideal gas, $C_v = (5/2)R$. Find the value of γ for this gas.

- 18.** A gas does 150 J of work while its internal energy decreases by 50 J. Find the heat supplied to (or removed from) the gas.
- 19.** Explain why the efficiency of a real heat engine is always less than that of a Carnot engine operating between the same two temperatures.
- 20.** A Carnot engine has an efficiency of 40% when the sink temperature is 300 K. Find the source temperature.
- 21.** State two limitations that make a Carnot engine an idealized, not practically achievable, engine.
- 22.** One mole of an ideal monatomic gas is heated at constant volume so that its temperature rises by 50 K. Find the heat absorbed ($R = 8.31 \text{ J/mol K}$).

Answer Key

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|---|---|
| 1. $\Delta U = 300 \text{ J}$ | 2. $W = 2740 \text{ J}$ (approx, using $nRT \ln 3$) |
| 3. $\eta = 40\%$ | 4. Conceptual - $W = 0$ since $\Delta V = 0$ |
| 5. $W = 200 \text{ J}$ | 6. Conceptual - ΔQ positive when heat is added to system; ΔW positive when work is done by the system |
| 7. $C_p = (5/2)R$, $\gamma = 5/3$ | 8. $\Delta U = 300 \text{ J}$ (increase) |
| 9. Conceptual - work done on the gas increases internal energy, raising temperature ($\Delta Q = 0$) | 10. $W = 400 \text{ J}$ |
| 11. Conceptual - no engine can convert heat completely into work with 100% efficiency | 12. Conceptual - second law forbids spontaneous heat flow from cold to hot; refrigerator needs work input to force this flow |
| 13. $\text{COP} = 6.5$ | 14. Conceptual - isothermal: T constant, slower process; adiabatic: $Q=0$, faster, no heat exchange |
| 15. $W = 600 \text{ J}$ | 16. Conceptual - e.g., quasi-static isothermal expansion of an ideal gas |
| 17. $\gamma = 7/5 = 1.4$ | 18. $\Delta Q = 100 \text{ J}$ (heat supplied) |
| 19. Conceptual - real engines have friction, heat losses, and irreversibilities absent in the idealized Carnot cycle | 20. $T_1 = 500 \text{ K}$ |
| 21. Conceptual - requires frictionless, quasi-static, no-heat-loss processes; not achievable in practice | 22. $Q = 623.25 \text{ J}$ |

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.